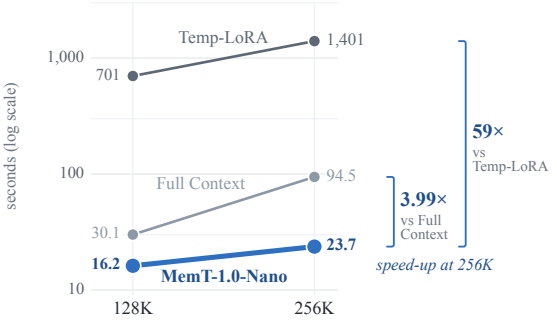
(a) End-to-end latency

history preparation + 128-token response



128K→256K latency: **MemT +46%** · Full Context +214% · Temp-LoRA +100%

(b) Persistent storage per trajectory

area to scale: ■ = 40 MiB (one MemT-1.0-Nano memory)

Full Context — *at 1M: too large*

5.061 GiB at 256K · ≈130 MemT memories

Temp-LoRA

172.5 MiB · 256K and 1M



MemT-1.0-Nano

40.00 MiB · 256K and 1M



130× less than Full Context (256K)

4.3× less than Temp-LoRA

fixed-size state: no growth with context